Prof. Stefan Minner
Principal Editor, *International Journal of Production Economics*

Resubmission of a substantially revised manuscript (previously PROECO-D-26-05160)
Replenishing perishables on shared delivery routes: Stock-age-based inventory policies, stochastic lookahead and decision-focused learning

Dear Professor Minner,

Thank you for your decision on PROECO-D-26-05160. You noted that the manuscript did not take a clear perishable inventory management perspective and therefore fell short in comparisons with state-of-the-art methods. I agree with both points. Rather than revising the text, I rebuilt the study around them; the new manuscript shares only the problem domain with the rejected version. I submit it as a new manuscript and summarise the changes below.

1. Perishable inventory management perspective.

- The problem is now a periodic-review perishable inventory system with a routing coupling: every day each policy observes the age-structured stock of every store, decides replenishment, and ships it on shared, capacitated routes; demand is served FIFO, unmet demand is lost and stock past its shelf life is discarded. The rejected version fixed a multi-day plan without observing stock, which is not how perishables are managed.
- Results are reported in inventory terms—cost composition, fill rate, waste rate, delivery frequency and store-level service—and analysed across three network layouts, shelf life, lost-sales/waste cost ratio, fleet tightness, routing cost, demand variability, issuing policy (FIFO vs. partial LIFO) and a case with observed fresh-food demand (FreshRetailNet-50K), with 100 test episodes per setting, three training seeds and prespecified, Holm-adjusted hypotheses.
- The structural results are stated in inventory terms and corrected: the recourse admits FIFO-optimal solutions once early disposal is allowed (the original lemma was false without it), the marginal value of a delivery is bounded by the lost-sales and waste penalties, and the decision-focused fixed point reduces to the perishable newsvendor fractile. All results are verified numerically.

2. Comparison with state-of-the-art methods.

- The comparator set now contains the perishable inventory rules recommended in the literature—periodic base-stock and base-stock with estimated waste (Broekmeulen and van Donselaar, 2009; Haijema and Minner, 2019)—the deliver-up-to-level rule of Crama et al. (2018) in its state-dependent form, a budgeted robust lookahead (Bertsimas and Thiele, 2006; Solyalı et al., 2012), and the Scenario Predict-then-Optimize method of Jia et al. (2025) with its published scenario set, together with mean and tuned-quantile lookahead and a perfect-information reference. All policies use the same information and the same forecasts and are tuned on separate validation data.
- Every lookahead decision is obtained from a MILP with exact route costs and integer routing on the implemented day, solved to a proven gap of at most 0.5%; the rejected version relied on heuristic fallbacks.
- The proposed method is now decision-focused learning applied on top of the strongest inventory rule: UL-DFL learns context- and stock-age-dependent order-up-to levels for the deliver-up-to-level rule from the realised cost of whole episodes. Decision-focused learning on top of the deterministic lookahead (LA-DFL) is retained and accompanied by an ablation with day-specific quantile levels.

- On fresh test episodes of the base case, decision-focused learning on top of the deliver-up-to-level rule (UL-DFL) is 1.1% cheaper than the rule re-tuned on a fine grid (Holm-adjusted $p < 0.001$) and wastes 28% less; across the thirteen settings it is significantly cheaper than the rule in nine, wastes less in twelve and is the cheapest policy in ten. On observed fresh-food demand the learned corrections did not transfer to new products, and the re-tuned rule was cheaper; we report this as a limitation. Decision-focused learning applied to the deterministic lookahead improves it, but a day-specific quantile level reproduces the gain, which we report as an ablation.

Scope and transparency. The synthetic demand is generated by a documented contextual model; the real-data case uses public data of one retailer, and its preparation (stock-out correction, rescaling to the network's capacities, exclusion of products with strong level shifts) is described in full. Because the synthetic test episodes had been used in an earlier round of the study, the prespecified primary hypotheses are tested on fresh test episodes of the base case; the deliver-up-to-level rule is tuned on a fine grid, and the first versions of its tuning and of the scenario-based comparator are reported in an appendix. Earlier claims about M5 calibration, an 81-cell factorial, regret against a heuristic clairvoyant plan and a solver speed-up have been withdrawn. Code, seeds and result files reproducing every number are provided.

Declarations. The manuscript is original, solely authored, and has not been published and is not under consideration elsewhere. I have no competing interests, and the work received no specific funding. The study involves no human participants or animal subjects. The use of generative AI (Claude, Anthropic) for code and text preparation is disclosed in the declarations file, together with the CRediT statement and the data and code availability statement. The manuscript file is anonymised; author details are given on the separate title page.

I hope that the revised study now meets the scope and standards of IJPE.

Sincerely,

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